



THE KING'S SCHOOL

2004 Higher School Certificate Trial Examination

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown for every question

Total marks – 84

- Attempt Questions 1-7
- All questions are of equal value
- Start a new booklet for each question
- Put your Student Number and the question number on the front of each booklet

Total marks – 84

Attempt Questions 1-7

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks) Use a SEPARATE writing booklet.

Marks

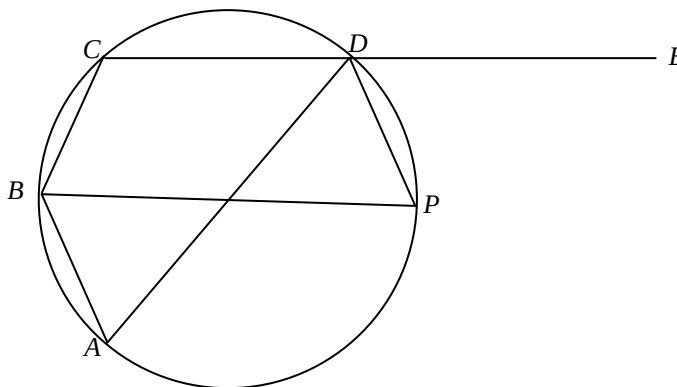
- (a) Find $\frac{d}{dx}(e^{\tan x})$. 2
- (b) The interval joining $A(x_1, y_1)$ and $B(x_2, y_2)$ is trisected by the points $(-2, 3)$ and $Q(1, 0)$. Write down the coordinates of A and B . 3
- (c) Find the acute angle, to the nearest degree, between the lines $x - y = 2$ and $2x + y = 1$. 2
- (d) Use the substitution $u = 1 - x$ to evaluate $3 \int_{-1}^0 \frac{x}{\sqrt{1-x}} dx$. 3
- (e) For a given series $T_{n+1} - T_n = 7$ and $T_1 = 3$. Find the value of S_{100} , where $S_n = T_1 + T_2 + \dots + T_n$. 2

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- (a) Solve $\frac{x^2 - 2}{x} < 1$. **3**
- (b) Find
- (i) $\int \frac{e^{2x}}{1 + e^{2x}} dx$. **1**
- (ii) $\int \frac{3}{5 + x^2} dx$. **2**
- (c) Solve the equation $2 \ln(3x + 1) - \ln(x + 1) = \ln(7x + 4)$. **3**
- (d) Solve $2 \tan^3 \theta - 3 \tan^2 \theta - 2 \tan \theta + 3 = 0$ for $0 \leq \theta \leq 360^\circ$, giving your answers to the nearest minute, where necessary. **3**

- (a) Use one application of Newton's Method to approximate the root of the equation $e^x + x = 2$ which is near 0.5, correct to two decimal places.

3

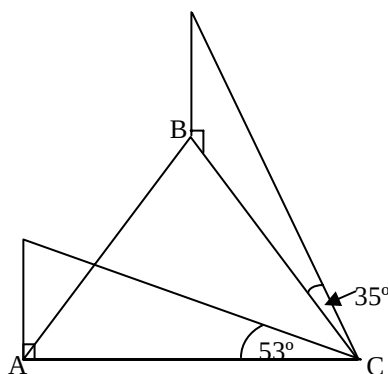
(b)



In the diagram above $ABCD$ is a cyclic quadrilateral. CD is produced to E . P is a point on the circle through A, B, C, D such that $\angle ABP = \angle PBC$.

- (i) Copy the diagram showing the above information.
- (ii) Explain why $\angle ABP = \angle ADP$.
- (iii) Show that PD bisects $\angle ADE$.
- (iv) If $\angle BAP = 90^\circ$ and $\angle APD = 90^\circ$, explain where the centre of the circle is located.
- (c) (i) Write $\cos x - \sqrt{3} \sin x$ in the form $A \cos(x + \alpha)$ where $A > 0$, $0 < \alpha < \pi$.
- (ii) Hence or otherwise, solve $\cos x - \sqrt{3} \sin x = 1$ for all values of x .

- (a) (i) Find the polynomial $P(x)$, if $P(x)$ has
- (α) degree 4;
 - (β) factors of $(x+3)^2$ and $(x-3)^2$; and
 - (γ) a remainder of -50 when divided by $x+2$.
- 2**
- (ii) Sketch the curve.
- 1**
- (b) The speed v cm/sec of a particle moving with simple harmonic motion in a straight line is given by $v^2 = 6 + 4x - 2x^2$, where x cm is the magnitude of the displacement from a fixed point O.
- (i) Show $\frac{d^2x}{dt^2} = -2(x-1)$.
- 2**
- (ii) Find the period of the motion.
- 1**
- (iii) Find the amplitude of the motion.
- 2**
- (c) A and B are the feet of two towers of equal height. B lies due North of A. From a point C, 40m East of A and in the same horizontal plane, the angle of elevation of the top of the tower A is 53° . From the same point the angle of elevation of tower B is 35° . Find the distance between the towers, AB, correct to the nearest metre.
- 4**



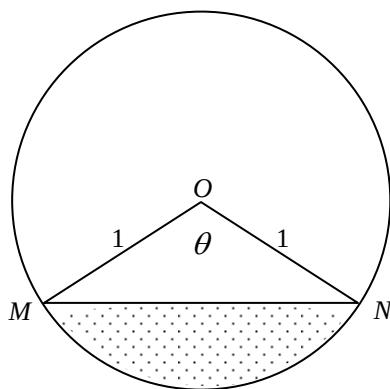
(a) For the function $y = 3\cos^{-1}\frac{x}{2}$

(i) Find the domain and the range. 2

(ii) Sketch the curve. 1

(iii) Find the equation of the tangent to the curve at the point on the curve where $x = 0$. 3

(b) O is the centre of a circle, radius 1m and $\angle MON = \theta$ radians. The shaded segment formed by MN has an area A square metres and perimeter P metres.



(i) Prove $A = \frac{1}{2}(\theta - \sin \theta)$. 1

and $P = \theta + 2 \sin \frac{\theta}{2}$. 1

(ii) P is increasing at a constant rate of R m/s. Find, in terms of R , the rate of increase of

(α) θ when $\angle MON = \frac{2\pi}{3}$; and 2

(β) A when $\angle MON = \frac{2\pi}{3}$. 2

-
- (a) Find the coordinates of the focus and the equation of the directrix of the parabola $x^2 = 4(x + y)$. 2
- (b) Prove by Induction that $3^{3n} + 2^{n+2}$ is divisible by 5 for all positive integers n . 4
- (c) Consider the variable point $T(-2t, t^2)$ on the parabola $y = \frac{1}{4}x^2$.
- (i) Prove that the equation of the tangent at T is $y + tx + t^2 = 0$ 2
- (ii) If A is the x intercept of the tangent at T , find the coordinates of A . 1
- (iii) Find the coordinates of M , the midpoint of the interval TA . 1
- (iv) Find the equation in Cartesian form of the locus of the point M given in part (iii). 2

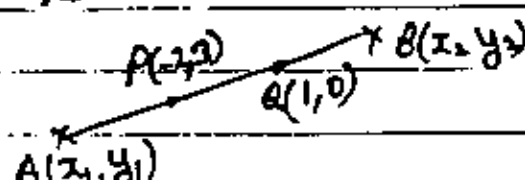
- (a) A stone is thrown from the top of a building 15m high with an initial velocity of 26 m/s at an angle of $\tan^{-1} \frac{5}{12}$ to the horizontal.

If the acceleration due to gravity is 10m/sec^2 , find

- | | | |
|-------|---|---|
| (i) | the greatest height above the ground reached by the stone | 2 |
| (ii) | the time of flight | 2 |
| (iii) | the range of the stone | 1 |
| (iv) | the velocity after 2 seconds | 2 |
- (b) Two of the roots of the equation $x^3 + ax^2 + b = 0$ are reciprocals of each other where a and b are real numbers.
- Show that
- | | | |
|-------|--|---|
| (i) | the third root is equal to $-b$; | 1 |
| (ii) | $a = b - \frac{1}{b}$; and | 2 |
| (iii) | the two roots, which are reciprocals, will be real if $-\frac{1}{2} \leq b \leq \frac{1}{2}$. | 2 |

End of Examination

1. a) $\sec x \cdot e^{\tan x}$

b) 

$$A: -2 = \frac{x_1 + 1}{2}; 3 = \frac{y_1 + 0}{2}$$

$$x_1 = -5; y_1 = 6$$

$$A(-5, 6)$$

$$B: 1 = \frac{-2 + x_2}{2}; 0 = \frac{3 + y_2}{2}$$

$$x_2 = 4; y_2 = -3$$

$$B(4, -3)$$

c) $l_1: m_1 = 1$

$$l_2: m_2 = -2$$

$$\tan \theta = \left| \frac{1 + 2}{1 + 1 \times -2} \right|$$

$$= 3$$

$$\theta = 72^\circ$$

e) $T_{n+1} - T_n = 7$

$$T_1 = 3$$

$$T_2 - T_1 = 7$$

$$\therefore T_2 = 10$$

$$T_3 - T_2 = 7$$

$$T_3 = 17$$

$$T_4 - T_3 = 7$$

$$T_4 = 24$$

$$S_n = 3 + 10 + 17 + \dots + T_n$$

$$\text{AS } a = 3, d = 7$$

$$S_{100} = \frac{100}{2} [2 \times 3 + 99 \times 7]$$

$$= 50 (6 + 693)$$

$$= 699 \times 50$$

$$= 34950$$

d) $u = 1 - x \quad x = -1, u = 2$

$$\frac{du}{dx} = -1 \quad x = 0, u = 1$$

$$I = 3 \int_2^1 \frac{1-u}{\sqrt{u}} \cdot -du$$

$$= 3 \int_2^1 \frac{u-1}{u^{\frac{1}{2}}} du$$

$$= 3 \int_2^1 u^{\frac{1}{2}} - u^{-\frac{1}{2}} du$$

$$= 3 \left[\frac{2}{3} u^{\frac{3}{2}} - 2u^{\frac{1}{2}} \right]_2^1$$

$$= 3 \left[\frac{2}{3} - 2 - \left(\frac{2}{3} \times \sqrt{8} - 2\sqrt{2} \right) \right]$$

$$= 3 \left[-\frac{4}{3} - \left(\frac{4}{3}\sqrt{2} - 2\sqrt{2} \right) \right]$$

$$= -4 - 4\sqrt{2} + 6\sqrt{2}$$

$$= 2\sqrt{2} - 4$$

Q2@ $\frac{x^2-2}{x} < 1$

c. Values: $x=0$

$$x^2-2=x$$

$$x^2-x-2=0$$

$$(x-2)(x+1)=0$$

$$x=2 \text{ or } -1$$

$$\begin{array}{c} + \quad + \quad + \\ -1 \quad 0 \quad 2 \end{array}$$

test $x = -\frac{1}{2} : \frac{\frac{1}{4}-2}{-\frac{1}{2}} = \frac{-1\frac{3}{4}}{-\frac{1}{2}} = 3\frac{1}{2}$ false

$x=1 : \frac{1-2}{1} < 1$ true

$\therefore x < -1 \text{ or } 0 < x < 2$

(b) (i)

$$\int \frac{2x}{1+e^{2x}} dx$$

$$= \frac{1}{2} \int \frac{2e^{2x}}{1+e^{2x}} dx$$

$$= \frac{1}{2} \ln(1+e^{2x}) + C$$

(ii)

$$\int \frac{3}{5+x^2} dx$$

$$= 3 \int \frac{1}{5+x^2} dx$$

$$= \frac{3}{\sqrt{5}} \tan^{-1} \frac{x}{\sqrt{5}} + C$$

(c) $2 \ln(3x+1) - \ln(x+1) = \ln(7x+4)$

$$\therefore \frac{(3x+1)^2}{x+1} = 7x+4$$

$$9x^2+6x+1 = 7x^2+11x+4$$

$$2x^2-5x-3 = 0$$

$$(2x+1)(x-3)=0$$

$$\therefore x = -\frac{1}{2} \text{ or } 3$$

but $3x+1 > 0$ & $x+1 > 0$ & $7x+4 > 0$

$$x > -\frac{1}{3} \text{ & } x > -1 \text{ & } x > -\frac{4}{7}$$

$$\therefore x > -\frac{1}{3}$$

$\therefore$ Solution $x = 3$

(d) $2 \tan^3 \theta - 3 \tan \theta - 2 \tan \theta + 3 =$

$$\tan^2 \theta (2 \tan \theta - 3) - 1(2 \tan \theta - 3)$$

$$\therefore (2 \tan \theta - 3)(\tan^2 \theta - 1) = 0$$

$$\therefore \tan \theta = \frac{3}{2} \text{ or } \pm 1$$

$$\theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ$$

$$56^\circ 19', 236^\circ 19'$$

Q3 (a) $y = e^x + x - 2$
 $\frac{dy}{dx} = e^x + 1$

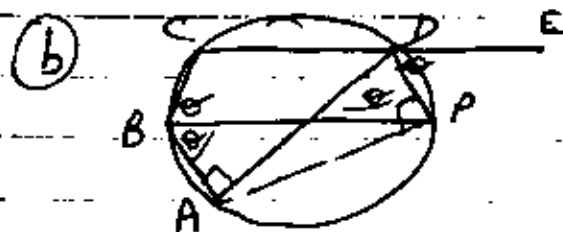
Let $x_1 = 0.5$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 0.5 - \frac{e^{0.5} + 0.5 - 2}{e^{0.5} + 1}$$

$$= 0.5 - \frac{0.48}{2.648}$$

$$= 0.44$$



Let $\angle ABP = \angle PBC = \theta$

(ii) $\angle ABP = \angle ADP = \theta$ (angles in same segment)

(iii) $\angle EDP = \angle PBC = \theta$ (ext. angle of cyclic quad = int. opp)

but $\angle ADP = \theta$ (from (ii))

$\therefore$ PD bisects $\angle ADE$

(iv) BP and AD are both diameters (angles in semi circle = 90°)
 $\therefore$ centre of circle is the point of intersection of BP & AD.

(c) (i) $\cos x - \sqrt{3} \sin x \equiv A \cos(x + \alpha)$
 $\equiv A [\cos x \cos \alpha - \sin x \sin \alpha]$

$A \cos \alpha = 1, A \sin \alpha = \sqrt{3}$

$A = \sqrt{1^2 + (\sqrt{3})^2} = 2$

$\sin \alpha = \frac{1}{2}$

$\alpha = \frac{\pi}{3}$

$\therefore \cos x - \sqrt{3} \sin x = 2 \cos(x + \frac{\pi}{3})$

$\therefore 2 \cos(x + \frac{\pi}{3}) = 1$

$\cos(x + \frac{\pi}{3}) = \frac{1}{2}$

$x + \frac{\pi}{3} = 2n\pi \pm \cos^{-1}(\frac{1}{2})$

$= 2n\pi \pm \frac{\pi}{3}$

$\therefore x = 2n\pi + 2\pi - \frac{\pi}{3}$

Q 4 (i) $P(x) = a(x+3)^2(x-3)^2$

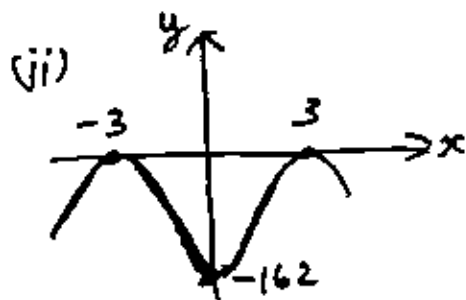
$P(-2) = -50$

$\therefore -50 = a(1)^2(-5)^2$

$-50 = 25a$

$a = -2$

$P(x) = -2(x+3)^2(x-3)^2$



(b) i) $v^2 = 6 + 4x - 2x^2$
 $\frac{d^2x}{dt^2} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$
 $= \frac{d}{dx} (3 + 2x - x^2)$

$= 2 - 2x$

$= -2(x-1)$

(ii) $\ddot{x} = -\pi^2 x$

$\pi = \sqrt{2}$

$\therefore \text{Period} = \frac{2\pi}{\sqrt{2}} = \sqrt{2}\pi \text{ sec.}$

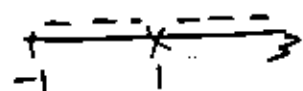
(iii) $v = 0$

$\therefore 2x^2 - 4x - 6 = 0$

$x^2 - 2x - 3 = 0$

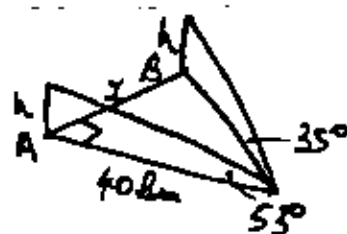
$(x-3)(x+1) = 0$

$\therefore x = -1 \text{ or } 3$



Amplitude = 2 cm.

(c)



$\tan 53^\circ = \frac{h}{40}$
 $h = 40 \tan 53^\circ$

$\tan 35^\circ = \frac{h}{BC}$
 $BC = \frac{h}{\tan 35^\circ}$

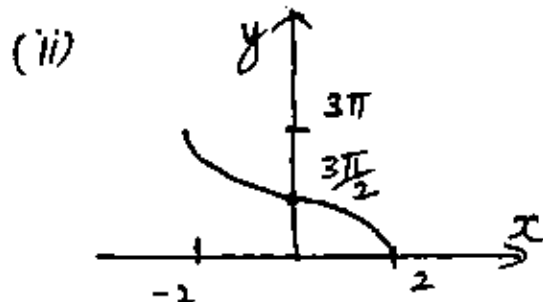
$\therefore BC = \frac{40 \tan 53^\circ}{\tan 35^\circ}$

$BC^2 = 2^2 + 40^2$
 $= \left(\frac{40 \tan 53^\circ}{\tan 35^\circ} \right)^2 + 40$
 $= 4146.94 \dots$
 $x = 63.396 \dots$
 $= 64 \text{ m.}$

Q5 @ $y = 3\cos^{-1}\frac{x}{2}$

(i) D: $-1 \leq \frac{x}{2} \leq 1$
 $-2 \leq x \leq 2$

R: $0 \leq y \leq 3\pi$
 $0 \leq y \leq 3\pi$



(iii) $y = 3\cos^{-1}\frac{x}{2}$
 $y' = \frac{-3}{\sqrt{1-\frac{x^2}{4}}} \times \frac{1}{2}$
 at $x=0$, $y' = \frac{-3}{2\sqrt{1}} = -\frac{3}{2}$
 $y = \frac{3\pi}{2}$

eqn of tangent is
 $y - \frac{3\pi}{2} = -\frac{3}{2}(x-0)$
 $2y - 3\pi = -3x$
 $3x + 2y - 3\pi = 0$

(b) (i) $A = \frac{1}{2} \times 1 \times 1 \times \theta - \frac{1}{2} \times 1 \times 1 \times \sin \theta$
 $= \frac{1}{2}(\theta - \sin \theta)$
 $P = 1 \times \theta + 2 \times \sin \frac{\theta}{2}$

(ii) $R = \frac{dP}{dt}$
 (a) $\frac{d\theta}{dt} = ?$

$P = \theta + 2\sin \frac{\theta}{2}$
 $\frac{dP}{dt} = 1 + \cos \frac{\theta}{2}$
 when $\theta = \frac{2\pi}{3}$, $\frac{dP}{dt} = 1 + \cos \frac{\pi}{3}$

$\frac{d\theta}{dt} = \frac{d\theta}{dP} \times \frac{dP}{dt}$
 $= \frac{2}{3} \times R$
 $= \frac{2R}{3} \text{ radians/sec}$

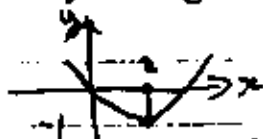
(B) $\frac{dA}{dt} = ?$
 $A = \frac{1}{2}(\theta - \sin \theta)$
 $\frac{dA}{d\theta} = \frac{1}{2}(1 - \cos \theta)$
 when $\theta = \frac{2\pi}{3}$, $\frac{dA}{d\theta} = \frac{1}{2}(1 - \cos \frac{2\pi}{3})$
 $= \frac{1}{2} \times \frac{3}{2}$

$\frac{dA}{dt} = \frac{dA}{d\theta} \times \frac{d\theta}{dt}$
 $= \frac{3}{4} \times \frac{2R}{3}$
 $= \frac{R}{2} \text{ m}^2/\text{sec}$

Q6) $x^2 = 4x + 4y$

$$x^2 - 4x + 4 = 4y + 4$$

$$(x-2)^2 = 4(y+1)$$



Focus (2, 0)

Directrix $y = -2$

⑦ When $n=1$, $3^{3n} + 2^{n+2} = 3^3 + 2^3 = 35$

which is divisible by 5
assume true for $n=k$

i.e. $\frac{3^{3k} + 2^{k+2}}{5} = c$ (an integer)

$$\therefore 3^{3k} = 5c - 2^{k+2}$$

Prove true for $n=k+1$

i.e. $\frac{3^{3k+3} + 2^{k+3}}{5} = c_1$ (an integer)

$$\text{LHS} = \frac{3^{3k} \cdot 3^3 + 2^{k+2} \cdot 2}{5}$$

$$= \frac{(5c - 2^{k+2}) \cdot 27 + 2^{k+2} \cdot 2}{5}$$

$$= \frac{5 \times 27c - 27 \times 2^{k+2} + 2 \times 2^{k+2}}{5}$$

$$= \frac{5[27c - 5 \cdot 2^{k+2}]}{5}$$

which is an integer if c is an integer

$\therefore$ if it is true for $n=k$, it is true for $n=k+1$
since it is true for $n=1$, it is true $n=2$
and so on for all n .



(i) $\frac{dy}{dx} = \frac{1}{2}x$

at T $\frac{dy}{dx} = -t$

eqn of tangent at T

$$y - t^2 = -t(x + 2t)$$

$$y - t^2 = -tx - 2t^2$$

$$y + tx + t^2 = 0$$

(ii) A $(-t, 0)$

(iii) M $(-\frac{3t}{2}, \frac{t^2}{2})$

(iv) $x = -\frac{3t}{2}$, $y = \frac{t^2}{2}$

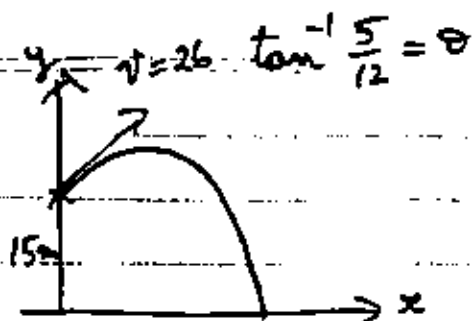
$$t = -\frac{2x}{3}$$

$$\therefore y = \frac{(-\frac{2x}{3})^2}{2}$$

$$= \frac{4x^2}{18}$$

$$y = \frac{2}{9}x^2$$

Q7 @



$$t=0, x=0, y=15, \dot{x} = 26 \times \frac{12}{13} = 24; \dot{y} = 26 \times \frac{5}{13} = 10$$

$$\ddot{y} = -10$$

$$\dot{y} = -10t + c_1$$

$$t=0, \dot{y} = 10, c_1 = 10$$

$$\dot{y} = -10t + 10$$

$$y = -5t^2 + 10t + c_2$$

$$t=0, y = 15, c_2 = 15$$

$$y = -5t^2 + 10t + 15$$

$$\ddot{x} = 0$$

$$\dot{x} = c_3 = 24$$

$$x = 24t + c_4$$

$$t=0, x=0, c_4 = 0$$

$$x = 24t$$

(i) greatest height $\dot{y} = 0, t = 1$

$$y = -5 + 10 + 15 = 20\text{m}$$

(ii) time of flight $y = 0$

$$t^2 - 2t - 3 = 0$$

$$(t+1)(t-3) = 0$$

$$\therefore t = -1 \text{ or } 3$$

Time of flight = 3 sec

(iii) when $t = 3$ $x = 24 \times 3$

$$= 72\text{m}$$

(iv) $t = 2$ $\dot{x} = 24, \dot{y} = 10 - 20 = -10$

$$v = \sqrt{24^2 + (-10)^2}$$

$$= 26\text{m per sec.}$$

$$Q7(b) \quad x^3 + ax^2 + b = 0$$

let roots be $\alpha, \frac{1}{\alpha}, \beta$

$$\therefore \alpha + \frac{1}{\alpha} + \beta = -a \quad \text{--- (1)}$$

$$\alpha \times \frac{1}{\alpha} + \alpha\beta + \frac{1}{\alpha} \times \beta = 0 \quad \text{--- (2)}$$

$$\alpha \times \frac{1}{\alpha} \times \beta = -b \quad \text{--- (3)}$$

$$(i) \quad \therefore \beta = -b \text{ from (3)}$$

$$(ii) \quad \alpha + \frac{1}{\alpha} - b = -a \text{ from (1)}$$

$$\alpha + \frac{1}{\alpha} = b - a$$

$$1 - \alpha b - \frac{b}{\alpha} = 0 \text{ from (2)}$$

$$1 - b(\alpha + \frac{1}{\alpha}) = 0$$

$$\therefore 1 - b(b - a) = 0$$

$$1 = b(b - a)$$

$$\frac{1}{b} = b - a$$

$$a = b - \frac{1}{b}$$

$$(iii) \quad \alpha + \frac{1}{\alpha} = b - a$$

$$= b - b + \frac{1}{b}$$

$$\alpha + \frac{1}{\alpha} = \frac{1}{b}$$

$$\alpha^2 - \frac{1}{b}\alpha + 1 = 0$$

real roots $\Delta \geq 0$

$$\therefore \frac{1}{b^2} - 4 \geq 0$$

$$1 - 4b^2 \geq 0$$

$$b^2 \leq \frac{1}{4}$$

$$-\frac{1}{2} \leq b \leq \frac{1}{2}$$